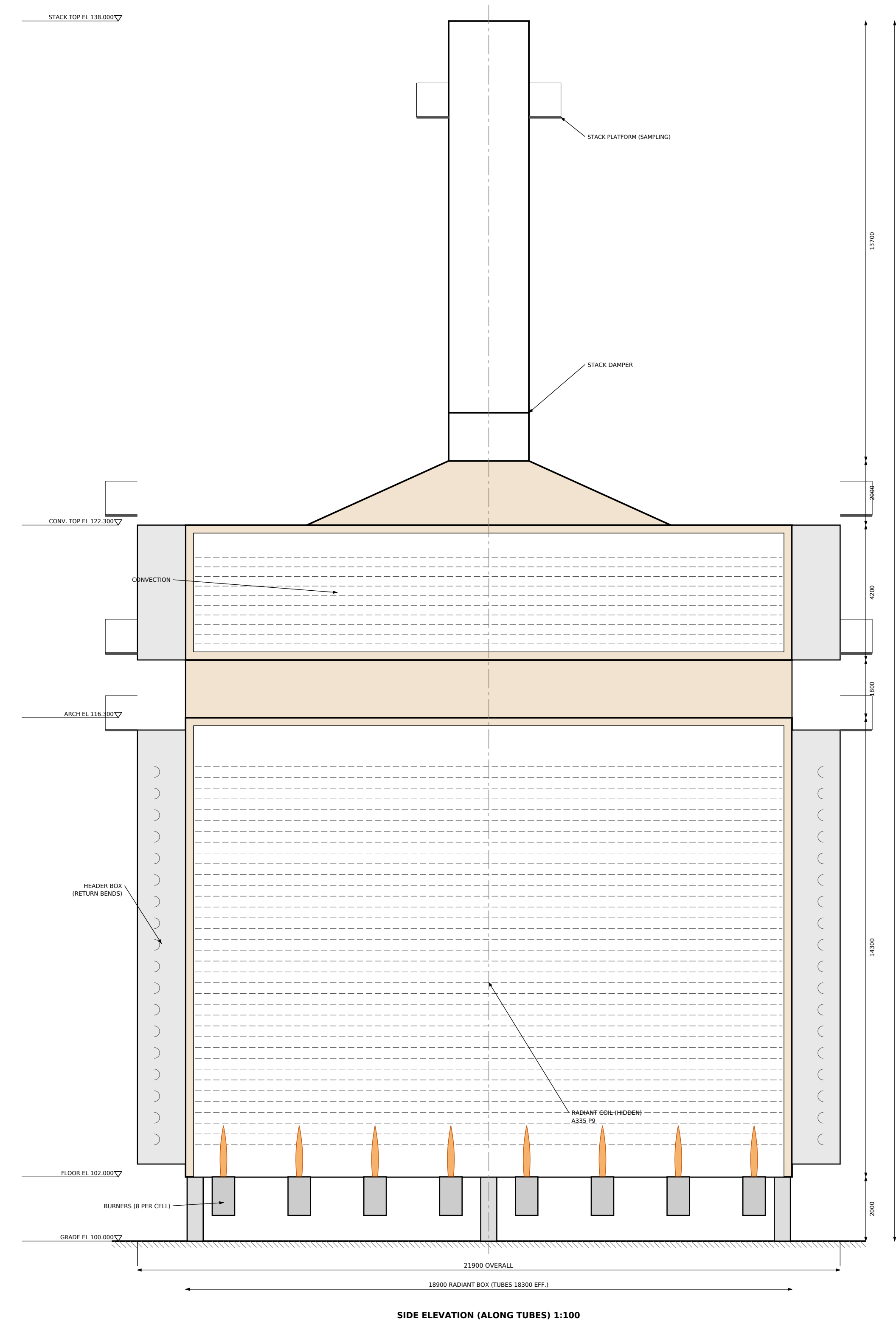
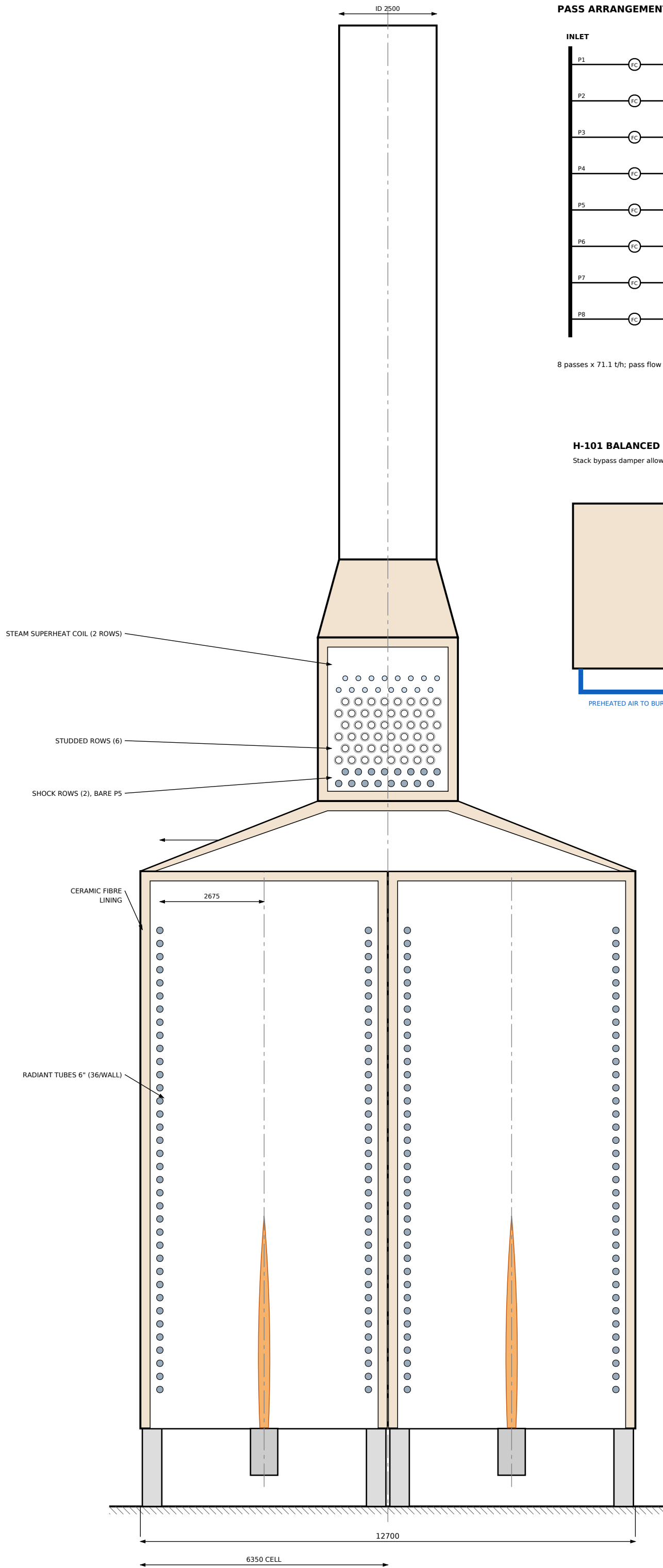


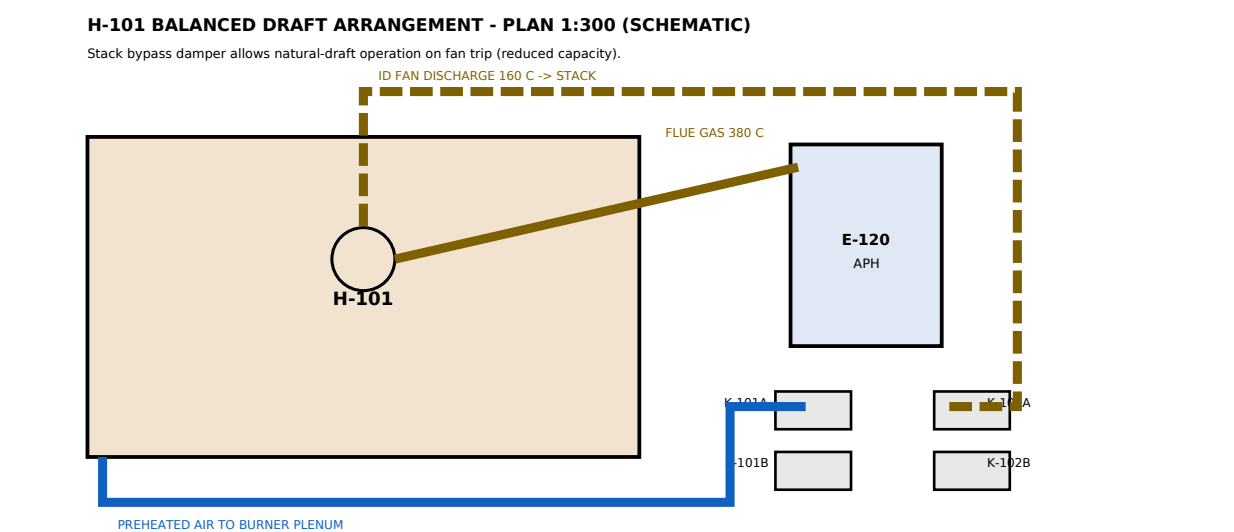
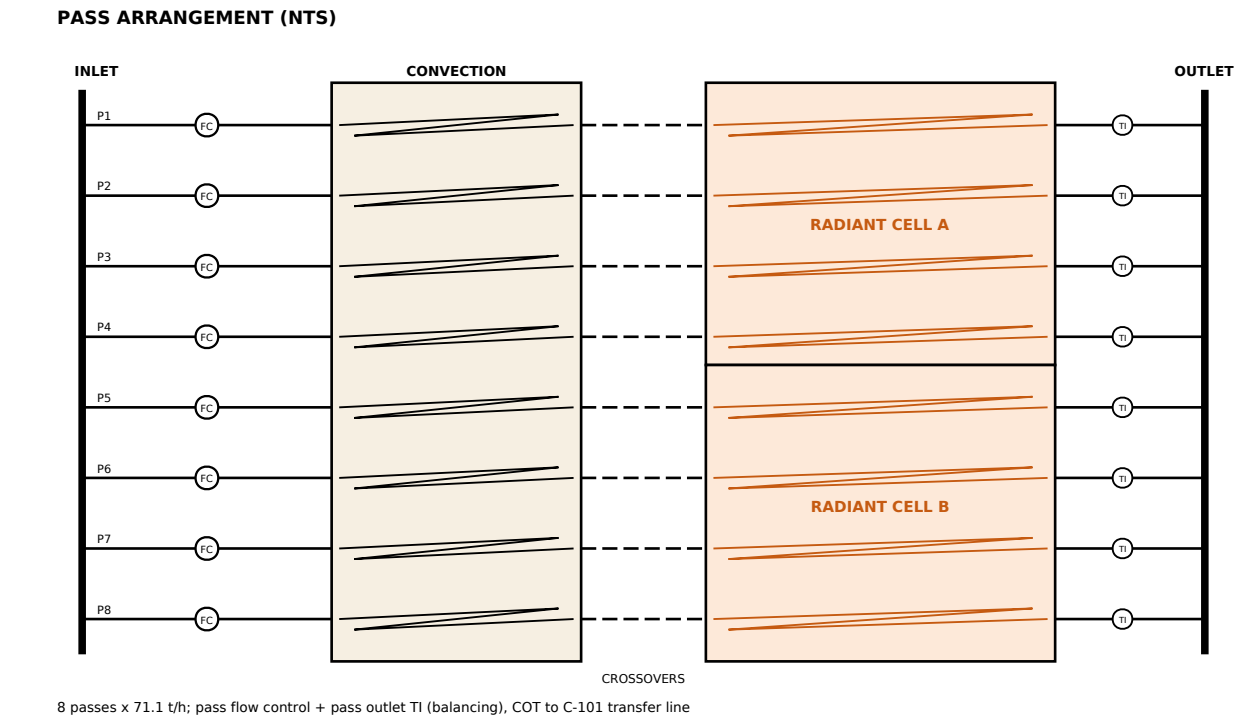
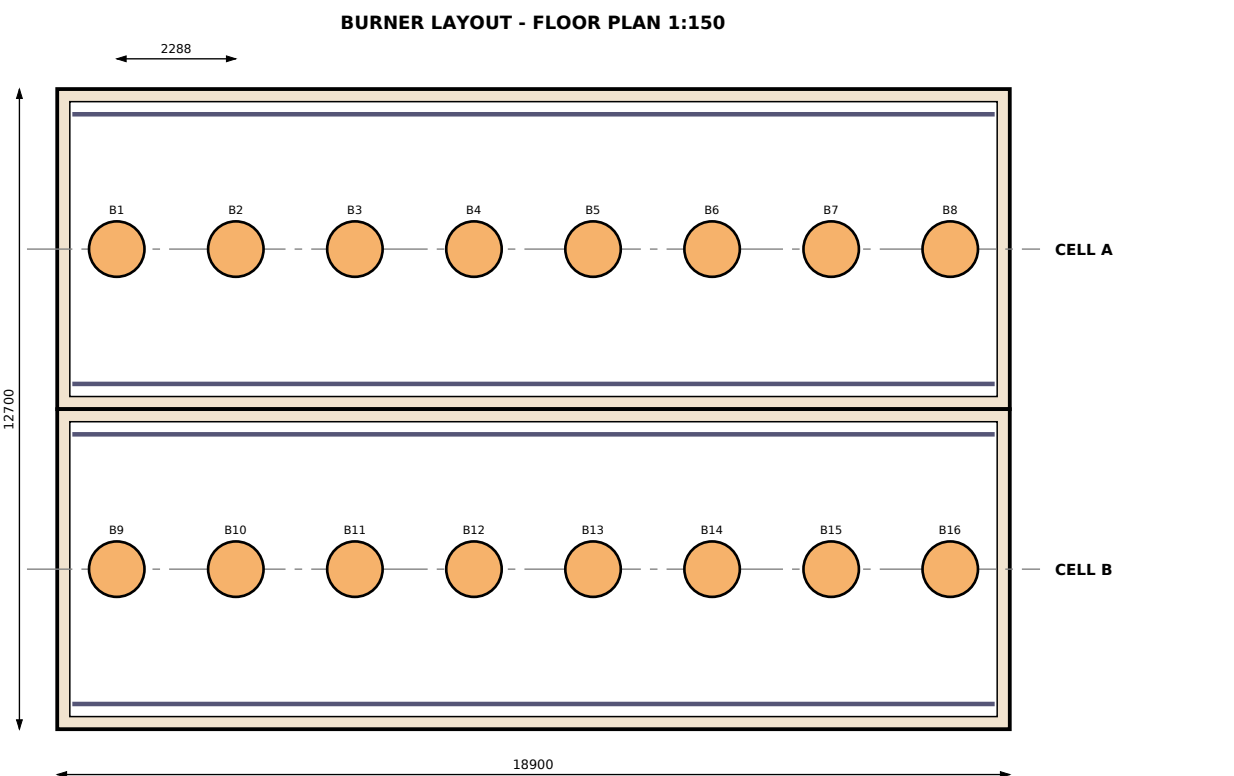
H-101 COIL SCHEDULE								
COIL SECTION	ROWS	TUBES/ROW	No. TUBES	OD x WALL mm	MATERIAL	EXT. SURFACE	DUTY MW	TDM C
Radiant (cells A+B)	36 wall	-	144	168.3 x 10.97	A335 P9	bare	41.7	510
Convection - shock	2	8	16	168.3 x 7.11	A335 P5	bare	-	-
Convection - studed	6	8	48	168.3 x 7.11	A106 B / P5	studs 12.7x25	17.9	-
Steam superheat (SS)	2	8	16	114.3 x 6.02	A335 P11	bare	1.13	-



SIDE ELEVATION (ALONG TUBES) 1:100



SECTION A-A (ACROSS CELLS) 1:100



H-101 DESIGN DATA	
Service	Atmospheric crude charge heater
Type	2-cell cabin, horizontal tubes, balanced draft + APH
Duty absorbed / fired	60.7 / 67.4 MW (LHV)
Radiant / convection duty	41.7 / 19.0 MW
Efficiency (LHV)	90 %
Flow / passes	569 t/h / 8
Inlet / outlet T	274 / 366 C
Outlet vaporisation	55 wt %
Avg. radiant flux	31.5 kW/m2
Mass flux	1176 kg/m2s
Bridgwall T (est.)	954 C
Flue gas	96 t/h
Overall L x W x H	21.9 x 12.7 x 38 m
Weight empty / operating	1077 / 1106 t

COIL DATA (API 530 - CFU-000-ME-CAL-001 S10)	
Radiant tubes	144 x 168.3 OD, Sch 80 (10.97 mm), A335 P9
Effective length	18.3 m, 36 per wall per cell
Design P elastic / rupture	38.5 / 17.0 barg
Design metal T (API 530)	510 C (EOR TMT 494 C)
T min elastic / rupture	6.12 / 3.51 mm (elastic governs)
Corr. allow. / life	3.0 mm / 100,000 h
Convection	64 tubes: 2 shock + 6 studed rows x 8
Extra coils	SS superheat 2 rows
Tube supports	HK40 radiant / SPCR-50Ni-Nb shock

BURNERS (API 535)	
Burners	16 ULNB staged-fuel gas, up-fired
Arrangement	single row per cell, pitch 2.29 m
Heat release design / max	4.23 / 5.27 MW
Burner CL to tube CL	2.67 m
Pilots / scanners	gas pilot, HEL, UV scanner each

- NOTES**
- Dimensions in mm, elevations in m; grade EL 100.000.
 - FEED arrangement - vendor to optimise (API 560).
 - Radiant floor 2.0 m above grade for burner access.
 - Tube pulling: removable header-box doors at both ends; crane access required.
 - Datasheet CFU-000-ME-DS-003; coil design CFU-000-ME-CAL-001 S10.
 - Balanced draft: FD fans K-101A/B, ID fans K-102A/B, APH E-120.
 - Estimated BWT 954 C - radiant surface to be reviewed (CAL-001 S13).

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
FIRED HEATER GENERAL ARRANGEMENT H-101 ATMOSPHERIC CRUDE CHARGE HEATER			
DISCIPLINE	STAGE	SCALE	SHEET
MECHANICAL	FEED	1:100	1 OF 1
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-ME-GA-005			A